

High-Gain Sub-3 dB Noise Figure Chip-Based Phase Sensitive Preamplifier with Pulsed Pumping

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Abstract We demonstrate a sub-3dB “black box” noise figure, Si_3N_4 waveguide-based phase-sensitive preamplifier receiver using pulsed pump. A 37 dB gain yields a sensitivity of -59 dBm at $\text{BER}=10^{-3}$ for a 2 Gb/s QPSK signal, with an estimated 0.5 dB on-chip noise figure. ©2026 The Author(s)

Introduction

To achieve high sensitivity, conventional coherent receivers typically employ erbium-doped fiber amplifiers (EDFAs) as preamplifier; however, their phase-insensitive amplification (PIA) imposes a 3 dB noise figure (NF) limit. Pulse position modulation (PPM) receivers can surpass this sensitivity limit but suffer from capacity restrictions imposed by single-photon detector bandwidth [1]. In contrast, phase-sensitive amplifiers (PSAs) can theoretically achieve 0 dB NF, offering a compelling route to enhanced receiver sensitivity [2].

Most existing $\chi^{(3)}$ nonlinearity-based PSAs use highly nonlinear fibers (HNLFs) and have demonstrated excellent sensitivity [2], [3]. However, the system typically requires hundreds-of-meters of HNLF, hindering integration. The watt-level pump power also induces stimulated Brillouin scattering (SBS), requiring pump phase modulation for mitigation [4]. Chip-based PSAs have recently emerged as a compact alternative, combining high nonlinearity, flexible dispersion engineering, and SBS-free operation.

A long waveguide is one of the key ingredients for realizing high performance PSAs. While meter length Si_3N_4 waveguides have been developed [5], reported net gains still remain at only 4.5 dB with NFs exceeding 3.7 dB, primarily limited by chip coupling loss, pump power, and chip thermal constraints. However, increasing waveguide length will eventually lead to an NF floor dictated by propagation loss. While recent chip-based parametric amplifiers have demonstrated ultra-broad gain bandwidths and opening new design avenues, it still offers limited improvement in net gain and NF [6], [7]. Compared with continuous-wave (CW) pumping, pulsed pumping enables significantly higher peak power at a fixed average power, simultaneously enhancing parametric gain and reducing NF without penalty. While this approach has been explored in parametric gain characterization [8], [9] it remains largely unexplored for communication applications.

In this paper, we demonstrate a below classical quantum limited NF chip-based PSA communication system using a pulsed pump with 13.6%

duty cycle and 2.4 W average power. With 2 Gbps QPSK signals, a fiber-to-fiber NF of 2.6 dB (back-to-back, B2B) and a PSA gain of 36.8 dB are achieved, where the on-chip NF is estimated to be ~0.5 dB after removing the ~2.1 dB input coupling loss. Benefiting from the high gain, PSA reception without EDFA cascading is also demonstrated. To the best of our knowledge, this represents the lowest fiber-to-fiber NF demonstrated in a chip-based $\chi^{(3)}$ PSA communication system to date.

Theoretical On Chip NF

In a lossy waveguide, the nonlinear interaction is governed by an effective length $L_{\text{eff}} = (1 - e^{-\alpha L})/\alpha$ with total transmission $A = e^{-\alpha L}$, where α is the linear loss and L the waveguide length. Under the undepleted-pump, small-signal, and near-phase-matched approximation, the PIA gain can be written as $G_{\text{PIA}} = \cosh^2(\gamma P_p L_{\text{eff}}) A$, where γ the nonlinear coefficient and P_p the on chip pump power [5]. When a phase-correlated idler is co-injected, coherent superposition via four-wave-mixing (FWM) yields the PSA gain $G_{\text{PSA}} = (\sqrt{G_{\text{PIA}}} + \sqrt{G_{\text{PIA}} - A})^2$, which approaches a 6 dB enhancement over G_{PIA} in the high-gain regime. The on-chip PSA NF can be expressed as:

$$NF_{\text{PSA}} = \frac{1}{2} \left[(1 + K) \left(1 + \frac{A}{G_{\text{PSA}}} \right)^2 - \frac{4K}{G_{\text{PSA}}} \right],$$

where $K = 1 - 2\ln A / (2\ln(G_{\text{PSA}} + A) - \ln(4G_{\text{PSA}}))$.

In practical systems, when the on-chip PSA gain is insufficient to meet the required input level of coherent receiver, an EDFA is normally cascaded after the chip for further amplification. The cascaded NF follows Friis' formula [10]:

$$NF_{\text{cascaded}} = NF_{\text{PSA}} + \frac{1/\eta_{\text{out}} - 1}{G'_{\text{PSA}}} + \frac{NF_{\text{EDFA}} - 1}{G'_{\text{PSA}} \eta_{\text{out}}},$$

where $G'_{\text{PSA}} = G_{\text{PSA}}/2$ refers to the per-wave gain, since the PSA input carries both signal and idler, and η_{out} is the passive transmission loss from the chip output to the EDFA input, including output coupling and filter loss before EDFA.

Current research to reduce chip-based PSA

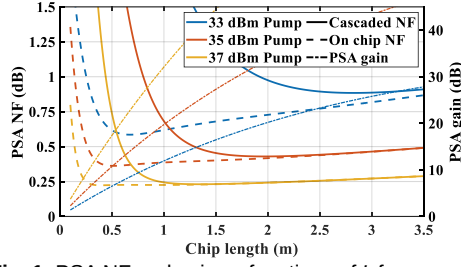


Fig. 1: PSA NF and gain as functions of L for a waveguide with $\gamma=0.9 \text{ W}^{-1}\text{m}^{-1}$, $\alpha=1.4 \text{ dB/m}$, $\eta_{\text{out}}=0.5$ under optimum phase-matching and ignoring input coupling loss.

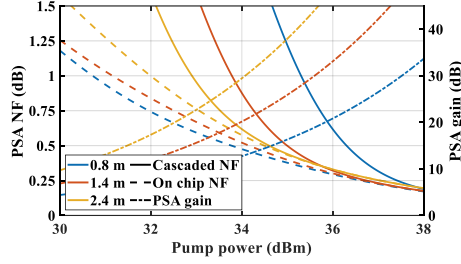


Fig. 2: PSA NF and gain as functions of pump power for a waveguide with $\gamma=0.9 \text{ W}^{-1}\text{m}^{-1}$, $\alpha=1.4 \text{ dB/m}$, $\eta_{\text{out}}=0.5$ under optimum phase-matching and ignoring input coupling loss.

NFs has mainly focused on increasing the waveguide length [5], [6]. However, Fig. 1 shows that the NF saturates once the waveguide is sufficiently long, indicating that extending the length alone eventually leads to an NF floor set by propagation loss. In contrast, Fig. 2 shows that increasing the pump power will continuously enhance both gain and NF, and similar NF values can be obtained for different chip lengths. Moreover, assuming a 4.5 dB cascaded EDFA NF, keeping the gap between the cascaded and on-chip NF below 0.1 dB requires a PSA gain of at least 26 dB. For instance, for a 1.4 m chip, although the on-chip NF can be as low as 1 dB, the cascaded NF remains very large when the PSA gain is low. This indicates that a low on-chip NF is insufficient for practical communication links unless accompanied by high PSA gain. Given the output power limit of commercial EDFAs and the power-handling constraint of the chip, pulsed pumping provides an effective route to further reduce the cascaded NF, enabling higher peak pump power under a certain average power.

Experimental setup

To experimentally investigate the proposed pulsed-pump PSA scheme, a chip-based setup was implemented as shown in Fig. 3. A 1.4-m-long Si_3N_4 waveguide, 1950 nm wide and 690 nm thick, with nonlinear coefficient of $0.96 \text{ W}^{-1}\text{m}^{-1}$, propagation loss of 1.42 dB/m and dispersion of $-28 \text{ ps}^2/\text{km}$ was used as the nonlinear medium [5]. Owing to fabrication asymmetry, the input and output facet coupling losses were approximately 2.1 dB and 3.5 dB, respectively. Since the chip supports multiple modes, intermodal coupling introduces strong wavelength-dependent loss, requiring careful selection of the operation wavelengths. Accordingly, the signal, pump, and idler were set to 1541.72 nm, 1553.91 nm and 1566.29 nm, respectively. This issue is largely mitigated in recent quasi-single-mode waveguides [6], [11].

At the transmitter, an external cavity laser (ECL) was modulated by an IQ modulator driven by an arbitrary waveform generator (AWG) to generate a 2 Gbps pulsed QPSK signal, as shown in Fig. 3(b). A CW laser (NKT E15, linewidth $<100 \text{ Hz}$) was amplified to 28.5 dBm and injected into a 114-m strained HNLF along with the signal to generate the idler. After filtering, a wave shaper (WS) was used to equalize the signal and idler powers, followed by a polarizer to remove the polarization distortion induced by the WS's polarization mode dispersion (PMD).

After equalization, the signal and idler were attenuated to the target levels via a variable optical attenuator (VOA). The pump for the receiver PSA was derived from an ultra-low-noise laser (Thorlabs ULN, linewidth $<100 \text{ Hz}$), modulated into the pulsed waveform by an IQ modulator driven by the same AWG as the transmitter (clock recovery would be required in a practical setup), and then phase-modulated by a dither tone for the phase-locking loop (PLL). After two-stage amplification to 33.8 dBm, the pump was co-injected with the signal into the nonlinear waveguide, yielding an on-chip pump average power of $\sim 31.7 \text{ dBm}$. As shown in Fig. 3(c), considering the 130 ps pulse width at 1 GHz repetition rate

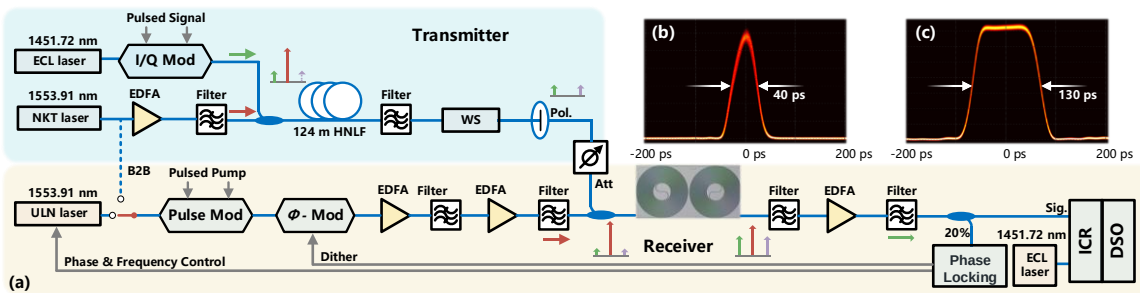


Fig. 3: (a) Experimental setup of the pulsed-pump PSA system. (b) Temporal waveform of the pulsed QPSK signal. (c) Temporal waveform of the pulsed pump.

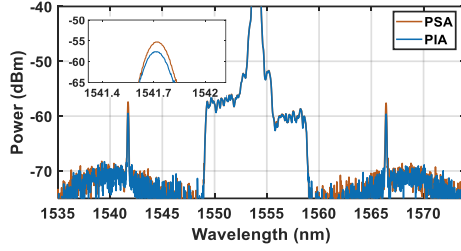


Fig. 4: Chip output spectra under PIA and PSA operation with a fixed input power (including signal and idler) of -50 dBm.

and the modulator extinction ratio, the pulsed waveform boosts the peak pump power by 8.4 dB above the average, reaching ~ 40.1 dBm. The PSA-amplified signal was filtered, optionally boosted by a second-stage EDFA, and detected by an integrated coherent receiver (ICR) with a 30 GHz real-time oscilloscope (DSO) for bit-error rate (BER) calculation. A PLL controller was used for laser frequency and phase stabilization, controlling the ULN laser current. In the “B2B configuration”, the NKT laser at the transmitter served as the receiver pump, eliminating most of the PLL associated penalty.

Experimental results

Figure 4 compares the chip output spectrum under PIA and PSA operation for a fixed input power of -50 dBm (including signal and idler) with a 0.1 nm resolution. The PSA provides 2.4 dB additional gain over the PIA, corresponding to the same optical signal-to-noise ratio (OSNR) improvement. While the 40.1 dBm peak pump power should ideally yield over 40 dB of PSA gain, the amplified spontaneous emission (ASE) spectra reveal that the signal and idler deviate from the optimum phase-matching point, reducing the measured PSA gain to ~ 36.8 dB. The intensity ripples in the ASE spectra stem from the limited optical spectrum analyser (OSA) sensitivity at low power level combined with the wavelength-dependent loss of the multimode waveguide.

Figure 5 compares the BER performance of a 2 Gbps QPSK signal amplified by the PSA, PIA, and a commercial EDFA as a function of received optical power (ROP) and corresponding received photons per bit (PPB), together with the theoretical BER curves for different NFs. The measured BERs agree well with the NF limited predictions, yielding fiber-to-fiber NFs of 2.6 dB, 3.0 dB, 5.4 dB, and 4.9 dB for the PSA (B2B), PSA, PIA, and EDFA, respectively. Notably, the 2.6 dB fiber-to-fiber NF under B2B surpasses the 3 dB classical quantum limit of PIA based receivers. After removing the ~ 2.1 dB input coupling loss, the on-chip NF is ~ 0.5 dB, within ~ 0.3 dB from the theoretical limit (Fig. 2). With separate transmitter and receiver lasers, the on-chip PSA NF increases to ~ 0.9 dB, while PIA on-chip NF is 3.3 dB. Since

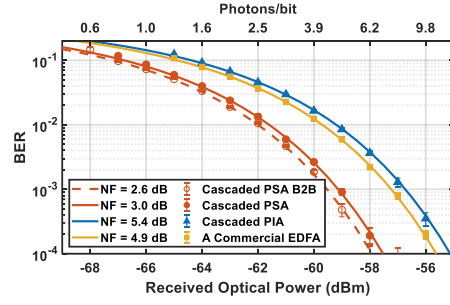


Fig. 5: BER (5 best in 10 measurements) versus ROP/PPB for a 2 Gbps QPSK signal amplified by the PSA and PIA (each cascaded with an EDFA) and EDFA, with theoretical NF-limited curves.

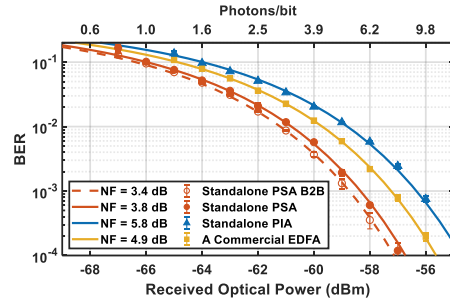


Fig. 6: BER (5 best in 10 measurements) versus ROP/PPB for a 2 Gbps QPSK signal amplified by the standalone PSA, PIA and EDFA, with theoretical NF-limited curves.

the chip and communication equipment are located in separate laboratories, the >100 m connecting fiber introduces a large loop delay that degrades PLL stability, resulting in the additional 0.4 dB PSA NF penalty [12]. This penalty is expected to deduce in a co-located implementation.

Benefiting from the ~ 36.8 dB gain of the pulsed-pump PSA, standalone reception without the cascaded EDFA is feasible. Fig. 6 shows the corresponding BER results, evaluated under the same conditions as Fig. 5. The estimated fiber-to-fiber NFs are 3.4 dB, 3.8 dB, 5.8 dB, and 4.9 dB for the PSA (B2B), PSA, PIA, and EDFA, respectively. Nevertheless, ROPs as low as -68 dBm still require a total gain of 50 dB to reach the ICR's penalty-free threshold, incurring a 0.4 dB PIA penalty in this measurement. For the PSA, this penalty increases to 0.8 dB, as the higher noise also impairs the monitor photodetector within PLL, degrading lock accuracy.

Conclusions

We have demonstrated a pulsed-pump Si_3N_4 PSA delivering 36.8 dB on-chip gain and ~ 0.5 dB on-chip NF, yielding 2.6 dB fiber-to-fiber NF with QPSK reception. The high gain also supports standalone operation without EDFA cascading. The current data rate is limited by the bandwidth of the AWG-driven pulse modulation, which can be increased by using dedicated pulsed lasers. These results establish a practical route toward deep-space coherent links.

Acknowledgements

This work was funded by the KA Wallenberg Foundation.

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